Solving “Infrequent Emojis” (50 pts – Crypto)

A step-by-step walkthrough of an emoji substitution cipher, including the exact method, tools, and websites used to crack it.

1. The Challenge

The challenge description read: “While DoomScrolling, our intel team came across a weird coded post with emojis all over it. One of our experts claims that the patterns point to it still being English but we don’t know what it means! Can you figure it out?” Submit the answer as `deadface{flag_text}`.

The key phrase is **“patterns point to it still being English.”** That is the giveaway: this is not a random image or steganography challenge — it is a classic substitution cipher, just using emoji characters as the cipher alphabet instead of letters or symbols.

2. The Cipher Image

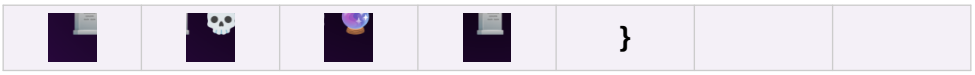


The original coded post, as found on DoomScroll. 30 symbols arranged in a 6-row grid, read left to right, top to bottom.

3. Transcribing the Cipher

The first step is careful transcription: write down every symbol in order, exactly as shown. Reading left to right, top to bottom (7 symbols per row), the cipher is:

	{					
	_				_	
		_			_	
		_				
		_				



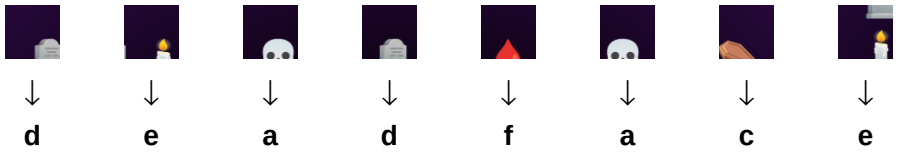
Note the underscores (word breaks) and curly braces — these are literal punctuation characters, not part of the cipher alphabet. They were left unencoded, giving a free hint about word structure.

4. Recognizing the Cipher Type

With underscores marking word boundaries and no repeated structure suggesting anything else, this is a **monoalphabetic substitution cipher**: each distinct emoji always maps to exactly one English letter, the same way a traditional "A = 1, B = 2" puzzle works, just with pictures standing in for letters. The task is to work out which emoji equals which letter.

5. The Key Insight: Known Plaintext

Rather than relying purely on frequency analysis (which needs a lot of text to be reliable, and this cipher is short), the fastest way in is **known plaintext**: every "deadface" CTF flag follows the exact format `deadface{...}`. Since the cipher includes a literal `{` character, the 8 symbols immediately before it *must* spell "deadface."



Matching these 8 symbols position-by-position against the letters of "deadface" immediately gives 6 mappings for free (two letters repeat):

Emoji	Letter	Emoji	Letter
	d		e
	a		f
	c		p
	l		s
	t		r
	y		o
	u		m
	n		w

6. Filling in the Rest with Pattern Matching

With over a dozen letters known, the rest of the message is solved by **pattern matching** — the classic word-puzzle technique of looking at a word's known/unknown letter pattern and guessing common English words that fit (the same technique used for Wordle or cryptogram puzzles).

Word 1 (after the brace): pattern `_ _ e a _ e` → fits "please".

Last word: pattern `s t a _ d a _ d` → fits "standard".

Remaining short words resolve from context: "try", "out", "my", "new".

7. The Decoded Message

```
please try out my new standard
```

8. The Flag

```
deadface{please_try_out_my_new_standard}
```

All 16 unique symbols map to 16 distinct letters with no conflicts — a clean one-to-one substitution, confirming the solve is correct.

9. General Method for This Cipher Type

Use this checklist any time you meet an emoji/symbol substitution cipher:

1. Transcribe carefully. Copy every symbol in exact reading order. Note punctuation (braces, underscores, spaces) separately — these are usually not part of the cipher and are free structural clues.
2. Look for a known-plaintext anchor. In a CTF, the flag format itself (e.g. deadface{...}, flag{...}, CTF{...}) is often visible in the ciphertext and instantly reveals several letters.
3. Use word-length and letter-pattern clues. A one-symbol word is almost always “a” or “I”. A repeated symbol inside a short word suggests double letters like “all”, “ee”, “oo”, “ss”.
4. Fill common short words first: “the”, “and”, “you”, “out”, “my”, “new” are frequent and easy to spot once a few letters are known.
5. Cross-check consistency. Every occurrence of the same symbol must decode to the same letter everywhere in the message — use this to catch mistakes.
6. If the message is long enough, classic letter-frequency analysis (E, T, A, O, I, N are the most common English letters) can substitute for or confirm a known-plaintext guess.

10. Tools & Websites That Help

Tool / Site	What it’s for
dcode.fr – Substitution Cipher solver	Automates monoalphabetic substitution solving via frequency analysis and pattern dict
quipqiup.com	A fast automatic cryptogram solver for substitution ciphers; useful for longer messages
CyberChef (gchq.github.io/CyberChef)	Useful for converting/transcribing emoji to code points, building frequency tables, and s
Emojipedia.org	Look up the exact name/meaning of an emoji if you’re unsure what symbol you’re look
Pen and paper / spreadsheet	For short ciphers like this one (30 symbols, 16 unique), manual pattern-matching by ha

Note: these are general-purpose, publicly available cipher/OSINT utilities and are not specific to this CTF. No proprietary or challenge-author-provided tooling was required — only careful transcription, the flag-format known-plaintext trick, and basic pattern matching.

11. Summary

Challenge	Infrequent Emojis (50 pts)
Cipher type	Monoalphabetic substitution (emoji → letter)
Key break-in	Flag prefix deadface{ gave 6 letters instantly
Technique	Known plaintext + pattern matching (no brute force needed)
Decoded message	please try out my new standard

Flag	deadface{please_try_out_my_new_standard}